

# Allelic Variation in Paralogs of GDP-L-Galactose Phosphorylase Is a Major Determinant of Vitamin C Concentrations in Apple Fruit<sup>1</sup>[C][W][OA]

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To identify the genetic factors underlying the regulation of fruit vitamin C (L-ascorbic acid [AsA]) concentrations, quantitative trait loci (QTL) studies were carried out in an F1 progeny derived from a cross between the apple (*Malus × domestica*) cultivars Telamon and Braeburn over three years. QTL were identified for AsA, glutathione, total antioxidant activity in both flesh and skin tissues, and various quality traits, including flesh browning. Four regions on chromosomes 10, 11, 16, and 17 contained stable fruit AsA-QTL clusters. Mapping of AsA metabolic genes identified colocations between orthologs of GDP-L-galactose phosphorylase (GGP), dehydroascorbate reductase (DHAR), and nucleobase-ascorbate transporter within these QTL clusters. Of particular interest are the three paralogs of *MdGGP*, which all collocated within AsA-QTL clusters. Allelic variants of *MdGGP1* and *MdGGP3* derived from the cultivar Braeburn parent were also consistently associated with higher fruit total AsA concentrations both within the mapping population (up to 10-fold) and across a range of commercial apple germplasm (up to 6-fold). Striking differences in the expression of the cv Braeburn *MdGGP1* allele between fruit from high- and low-AsA genotypes clearly indicate a key role for *MdGGP1* in the regulation of fruit AsA concentrations, and this *MdGGP* allele-specific single-nucleotide polymorphism marker represents an excellent candidate for directed breeding for enhanced fruit AsA concentrations. Interestingly, colocations were also found between *MdDHAR3-3* and a stable QTL for browning in the cv Telamon parent, highlighting links between the redox status of the AsA pool and susceptibility to flesh browning.

In plants, L-ascorbic acid (AsA; vitamin C) is essential for the detoxification of reactive oxygen species produced under stress or following exposure to pathogens. In addition to these antioxidant functions, AsA has been shown to be involved in a range of important cellular processes, including plant development and hormone signaling, cell cycle, cell expansion, senescence, and as a cofactor for a number of important enzymes (for review, see Davey et al., 2000; Smirnoff et al., 2001; Noctor, 2006). Fruit AsA concentrations have also been correlated with the maintenance of quality during postharvest storage

(Davey and Keulemans, 2004; Davey et al., 2007) and have been linked to susceptibility to internal browning in both apple (*Malus × domestica*; Davey et al., 2006; Davey and Keulemans, 2009) and pear (*Pyrus communis*; Veltman et al., 1999; Franck et al., 2003). Finally, AsA is clearly an essential dietary component for humans, with a protective role proposed for many disorders and diseases (Diplock et al., 1998). Given its importance for all metabolically active tissues, there is widespread interest in unraveling the mechanisms underlying the genetic control of AsA concentrations in fruits as well as in how AsA interacts with other plant antioxidant pools.

The concentration of AsA will be determined by the net rates of biosynthesis, recycling, degradation, and/or intercellular and intracellular transport, but the relative contribution of these various processes depends on several factors, including genetics, tissue type (Bulley et al., 2009), developmental stage (Hancock et al., 2007; Bulley et al., 2009; Ioannidi et al., 2009), and light intensity (Yabuta et al., 2007; Gautier et al., 2009). The biosynthesis of AsA proceeds via L-Gal (Wheeler et al., 1998), although conclusive evidence for all steps has only relatively recently become available (Conklin et al., 2006; Laing et al., 2007). Alternative biosynthetic routes involving uronic acids (Davey et al., 1999; Agius et al., 2003), L-gulose (Wolucka and Van Montagu, 2003), or

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myoinositol (Lorence et al., 2004) have been proposed in several plant species, including apple (Davey et al., 2004; Razavi et al., 2005; Fig. 1), but their physiological relevance and contribution to the AsA pool is still far from clear in most plant species, with the possible exception of strawberry (*Fragaria × ananassa*; Agius et al., 2003; Cruz-Rus et al., 2011; Zorrilla-Fontanesi et al., 2011).

As an antioxidant, AsA is able to accept electrons from a wide range of radical substrates, and in this process it becomes oxidized first to monodehydroascorbate and then to dehydroascorbate (DHA). These oxidized forms of AsA can be regenerated by the ascorbate-glutathione (GSH) cycle, so that GSH and the activities of GSH reductase, dehydroascorbate reductase (DHAR), and monodehydroascorbate reductase (MDHAR) maintain the size and redox status of the AsA pool (Noctor and Foyer, 1998; Fig. 1). Indeed, overexpression of an *Arabidopsis* (*Arabidopsis thaliana*) MDHAR (Eltayeb et al., 2007) and a wheat (*Triticum aestivum*) DHAR (Chen et al., 2003) have both been shown to increase foliar AsA concentrations in tobacco (*Nicotiana tabacum*). MDHAR activity has also been positively correlated with both AsA and fruit firmness in tomato (*Solanum lycopersicum*) after chilling stress (Stevens et al., 2008).

Tissue AsA concentrations can also be maintained by intercellular transport, and there is evidence for the long-distance transport of AsA via the phloem from source (leaf) to sink (fruit) tissues (Franceschi and Tarlyn, 2002; Hancock et al., 2003). In apple, fruit AsA concentrations have been suggested to be partly dependent on the translocation of AsA from leaves (Li et al., 2009), but in black currant (*Ribes nigrum*), others concluded that the contribution of phloem AsA transport to fruit AsA concentrations was negligible (Hancock et al., 2007). While the actual mechanisms of long-distance transport of AsA have not been fully determined, attention has focused on the large family of Nucleobase-Ascorbate Transporters (NATs; de Koning and Djalilinas, 2000), and NAT homologs have been found to be highly expressed in vascular tissues (Maurino et al., 2006).

Genes involved in several of these mechanisms have been proposed to be key regulators of fruit AsA concentrations, including GDP-L-Gal phosphorylase (GPP) or *vitamin c defective2* (VTC2) in kiwifruit (*Actinidia deliciosa*; Bulley et al., 2009, 2012) as well as GDP-Man-3,5-epimerase (GME; Gilbert et al., 2009), L-Gal-1-P-phosphatase (GPP or VTC4; Ioannidi et al., 2009), and MDHAR (Stevens et al., 2007) in tomato. However, apart from GPP (Bulley et al., 2012), overexpression of these structural genes has to date had limited success in altering the fruit AsA pool (Agius et al., 2003; Bulley et al., 2009; Haraldsen et al., 2011).

In this work, we set out to identify potential genetic determinants of fruit AsA concentrations in apple fruit using a combination of molecular and genomic approaches. Initial quantitative trait loci (QTL) analyses of AsA concentrations (Davey et al., 2006) have been expanded to identify QTL for other antioxidants and fruit quality traits over three years, including

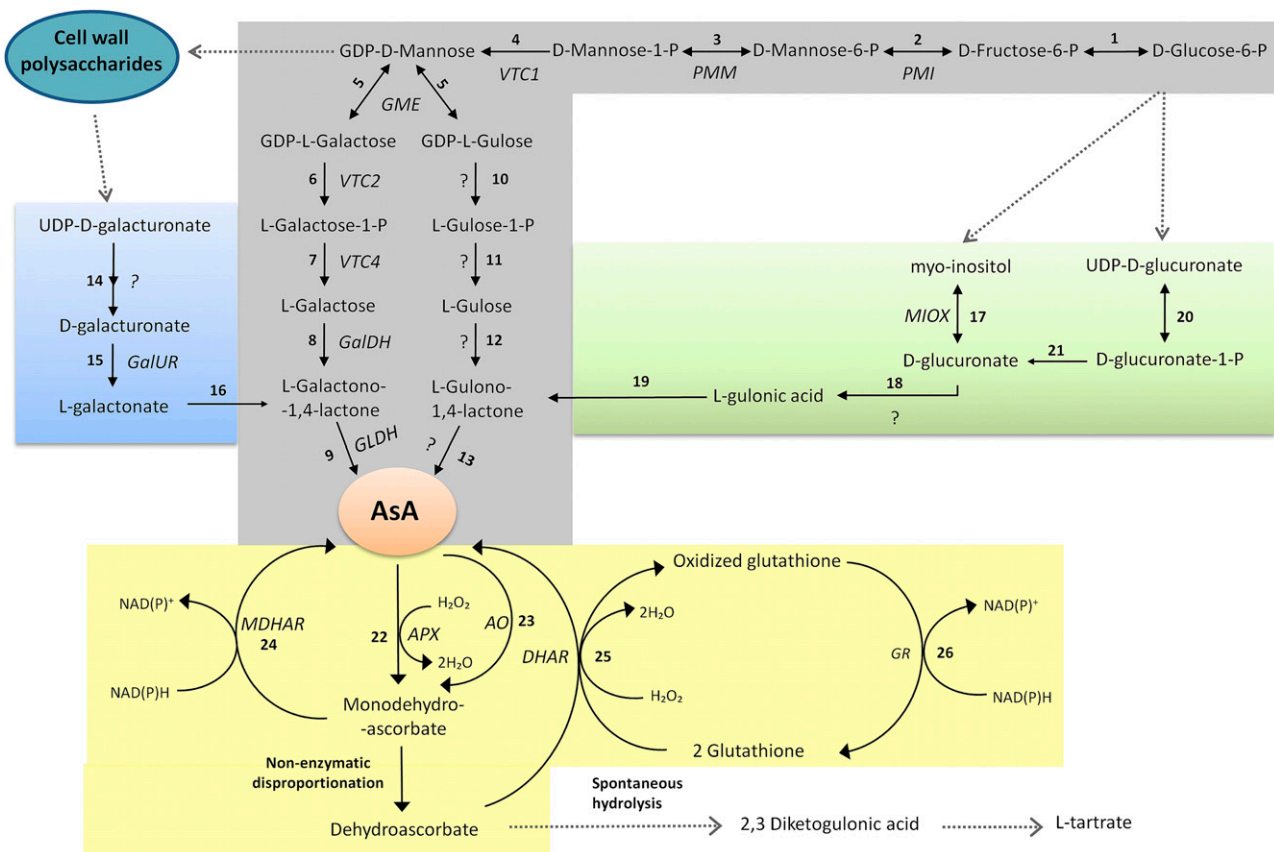
results in 1 year comparing the concentrations of AsA in fruit and leaves. Alignments of the apple orthologs of genes involved in AsA biosynthesis, turnover, and transport against the whole genome sequence of cv Golden Delicious (Velasco et al., 2010) allowed us to identify candidate genes (CGs) colocating with stable QTL clusters. Using next-generation sequencing (RNA-Seq) data, polymorphic single-nucleotide polymorphism (SNP)-based markers were developed for these colocating CGs, and their positions on individual linkage groups were confirmed by linkage mapping in our mapping population. Finally, associations between allelic variants of these CGs and their expression levels in cultivars with contrasting AsA concentrations allowed us to develop allele-specific markers associated with high fruit AsA concentrations.

## RESULTS

### Population Variation in Tissue AsA and GSH Concentrations

Fruit AsA-total ascorbic acid (totAsA) and GSH-total glutathione (totGSH) concentrations showed a high degree of variability in the cv Telamon × Braeburn (TxB) mapping population, with mean values for flesh totAsA and totGSH on average varying 10- and 7-fold across the population over the three years of measurements, respectively (Table I). Total antioxidant activity (TAA) was only measured in 2006, with results indicating that the contribution of totAsA to TAA ranged from 4% to 25% in flesh and from 6% to 50% in skin tissues. Both AsA and totAsA (AsA + DHA) and GSH and totGSH (GSH + oxidized glutathione [GSSG]) concentrations in flesh and skin tissues showed normal distribution across progeny of the cv TxB population over the years, but concentrations of the oxidized forms (DHA and GSSG) displayed a statistically normal distribution in only one of the three years of measurement (2006).

Fruit AsA, totAsA, GSH, and totGSH concentrations in individuals of the TxB progeny were positively correlated ( $P < 0.0001$ ) between flesh and skin tissues, such that fruit with high skin AsA and GSH concentrations also contained high flesh AsA and GSH concentrations (Supplemental Table S1). AsA and totAsA concentrations were also significantly correlated across the years (Supplemental Table S2;  $P < 0.0001$ ); however, correlations for DHA, GSH, GSSG, and totGSH concentrations were lower or not significant at all. Comparing totAsA concentrations across the years indicated that in 2009, values were 27% and 42% higher in fruit flesh and 40% and 60% higher in fruit skin than the concentrations determined in 2005 and 2006, respectively. The percentage proportion of DHA in the totAsA pool, which is an indicator of the degree of oxidative stress experienced by the tissue, was found to be significantly higher in 2006 compared with 2005 or 2009. Finally, fruit skin tissues contained approximately 40% more totGSH than flesh tissues, and values in both tissues were around 50% higher in 2009 than in 2006.



**Figure 1.** AsA biosynthetic and recycling pathways in plants: L-Gal pathway, reactions 1 to 9; L-gulose pathway, reactions 1 to 5 and 10 to 13; D-galacturonate pathway, reactions 14 to 16; myoinositol/glucuronate pathway, reactions 17 to 21; recycling pathway, reactions 22 to 27. Reactions with question marks are yet to be identified. Numbered reactions are as follows: 1, Glc-6-P isomerase; 2, Man-6-P isomerase (PMI; EC 5.3.1.8); 3, phosphomannomutase (PMM; EC 5.4.2.8); 4, GDP-D-Man pyrophosphorylase (GMP; EC 2.7.7.13); 5, GDP-D-Man 3',5'-epimerase (GME; EC 5.1.3.18); 6, GDP-L-Gal-phosphorylase (VTC2 or GGP; EC 2.7.7.69); 7, L-Gal-1-P phosphatase (VTC4 or GPP; EC 3.1.3.25); 8, L-Gal dehydrogenase (GalDH; EC 1.1.1.48); 9, L-galactono-1,4-lactone dehydrogenase (GLDH; EC 1.3.2.3); 10, nucleotide pyrophosphatase or sugar-1-phosphate guanylttransferase; 11, sugar phosphatase; 12, sugar dehydrogenase; 13, L-gulono-1,4-lactone oxidase (EC 1.1.3.8); 14, D-galacturonate-1-phosphate uridylyltransferase and D-galacturonate-1-phosphate phosphatase (possible); 15, D-galacturonate reductase (GalUR; EC 1.1.1.n9); 16, aldolactonase; 17, myoinositol oxygenase (MIOX; EC 1.13.99.1); 18, D-glucuronate reductase (EC 1.1.1.19); 19, L-gulonolactonase; 20, D-glucuronate-1-phosphate uridylyltransferase; 21, D-glucurono-1-phosphate phosphatase; 22, L-ascorbate peroxidase (APX; EC 1.11.1.11); 23, L-ascorbate oxidase (AO; EC 1.10.3.3); 24, monodehydroascorbate reductase (MDHAR; EC 1.6.5.4); 25, dehydroascorbate reductase (DHAR; EC 1.8.5.1); 26, GSH reductase (GR; EC 1.8.1.7). [See online article for color version of this figure.]

These year-to-year differences in fruit AsA and GSH traits are probably related to significant differences in the climatic conditions (Supplemental Fig. S1). For example, the length of the harvest season ranged from 32 d in 2006 to 59 d in 2005, even though the first day of the harvesting period was the same each year. The highest mean population AsA and totAsA concentrations were measured in 2009, when both average daily temperature and hours of sun radiation were higher than those of the other years, suggesting an adaption to the high light and high temperature experienced in this year. Similarly, the lowest AsA and totAsA concentrations were measured in 2006, when hours of sunlight were lower than in either 2005 or 2009. A comparison of leaf and fruit totAsA and totGSH concentrations in

2009 indicated that apple leaves contained approximately 65-fold more totAsA and 12-fold more totGSH than fruit flesh tissues (Supplemental Table S3). No significant correlations were found between fruit and foliar AsA and totAsA concentrations in fruit of progeny from TxB, but there were significant correlations between foliar and both fruit flesh and skin GSH, totGSH, and GSSG concentrations and percentage GSSG (Supplemental Table S4).

### Population Variation in Flesh Browning and Fruit Quality Traits

The susceptibility of fruit flesh to browning was assessed at harvest in 2006 and 2009 by measuring the

**Table 1.** Overview of population mean values and distribution for fruit antioxidant traits of flesh (FL) and skin (SK) tissues, as well as fruit fresh weight over the years

Fruit AsA, totAsA, DHA, and percentage DHA were measured over a period of three years (2005, 2006, and 2009); fruit GSH, totGSH, GSSG, and percentage GSSG were analyzed over 2 years (2006 and 2009); whereas fruit TAA was measured only in 2006. Each trait value represents the mean of the results from the analysis of 10 randomly chosen apples per *n* genotypes (2005, *n* = 136; 2006, *n* = 140; 2009, *n* = 163)  $\pm$  sd.

| Trait                                                | F1 Progeny |                   |        |         |              |
|------------------------------------------------------|------------|-------------------|--------|---------|--------------|
|                                                      | Year       | Mean $\pm$ sd     | Lowest | Highest | Distribution |
| AsA FL (nmol g <sup>-1</sup> fresh weight)           | 2005       | 438.5 $\pm$ 166.4 | 84.6   | 909.6   | Normal       |
|                                                      | 2006       | 313.5 $\pm$ 136.7 | 43.7   | 828.1   | Normal       |
|                                                      | 2009       | 600.2 $\pm$ 238.4 | 155.9  | 1,473   | Normal       |
| DHA FL (nmol g <sup>-1</sup> fresh weight)           | 2005       | 52.0 $\pm$ 75.4   | 0.0    | 426.3   | Nonnormal    |
|                                                      | 2006       | 68.2 $\pm$ 29.7   | 0.0    | 152.3   | Normal       |
|                                                      | 2009       | 55.3 $\pm$ 38.6   | 8.0    | 267.1   | Nonnormal    |
| Percentage DHA FL                                    | 2005       | 10.3 $\pm$ 11.8   | 0.0    | 61.1    | Nonnormal    |
|                                                      | 2006       | 19.2 $\pm$ 8.44   | 0.0    | 55.0    | Normal       |
|                                                      | 2009       | 8.4 $\pm$ 4.1     | 1.2    | 30.2    | Nonnormal    |
| totAsA FL (nmol g <sup>-1</sup> fresh weight)        | 2005       | 490.5 $\pm$ 176.7 | 111.8  | 1,034   | Normal       |
|                                                      | 2006       | 381.7 $\pm$ 148.4 | 97.2   | 973.0   | Normal       |
|                                                      | 2009       | 654.1 $\pm$ 256.5 | 164.8  | 1,663   | Normal       |
| AsA SK (nmol g <sup>-1</sup> fresh weight)           | 2005       | 1,617 $\pm$ 464.9 | 413.7  | 2,969   | Normal       |
|                                                      | 2006       | 1,077 $\pm$ 332.2 | 405.8  | 2,459   | Normal       |
|                                                      | 2009       | 2,705 $\pm$ 796.6 | 1,052  | 5,883   | Normal       |
| DHA SK (nmol g <sup>-1</sup> fresh weight)           | 2005       | 158.0 $\pm$ 188.0 | 0.0    | 1,262   | Nonnormal    |
|                                                      | 2006       | 115.2 $\pm$ 70.5  | 0.0    | 580.7   | Normal       |
|                                                      | 2009       | 242.0 $\pm$ 129.1 | 44.3   | 758.0   | Nonnormal    |
| Percentage DHA SK                                    | 2005       | 9.0 $\pm$ 9.5     | 0.0    | 46.0    | Nonnormal    |
|                                                      | 2006       | 9.9 $\pm$ 5.3     | 0.0    | 35.8    | Normal       |
|                                                      | 2009       | 8.8 $\pm$ 5.1     | 1.8    | 42.3    | Nonnormal    |
| totAsA SK (nmol g <sup>-1</sup> fresh weight)        | 2005       | 1,775 $\pm$ 470.6 | 465.4  | 3,377   | Normal       |
|                                                      | 2006       | 1,193 $\pm$ 350.3 | 464.1  | 2,660   | Normal       |
|                                                      | 2009       | 2,947 $\pm$ 819.8 | 1,289  | 6,260   | Normal       |
| GSH FL (nmol g <sup>-1</sup> fresh weight)           | 2005       | —                 | —      | —       | —            |
|                                                      | 2006       | 27.8 $\pm$ 11.1   | 8.0    | 81.4    | Nonnormal    |
|                                                      | 2009       | 60.8 $\pm$ 20.8   | 16.8   | 152.8   | Normal       |
| GSSG FL (nmol g <sup>-1</sup> fresh weight)          | 2005       | —                 | —      | —       | —            |
|                                                      | 2006       | 11.2 $\pm$ 5.3    | 0.0    | 27.6    | Normal       |
|                                                      | 2009       | 21.6 $\pm$ 14.7   | 1.5    | 84.3    | Nonnormal    |
| Percentage GSSG FL                                   | 2005       | —                 | —      | —       | —            |
|                                                      | 2006       | 28.9 $\pm$ 10.3   | 0.0    | 52.5    | Normal       |
|                                                      | 2009       | 25.5 $\pm$ 14.0   | 2.6    | 67.7    | Nonnormal    |
| totGSH FL (nmol g <sup>-1</sup> fresh weight)        | 2005       | —                 | —      | —       | —            |
|                                                      | 2006       | 39.0 $\pm$ 13.7   | 10.5   | 93.2    | Nonnormal    |
|                                                      | 2009       | 82.4 $\pm$ 26.2   | 34.4   | 184.0   | Nonnormal    |
| GSH SK (nmol g <sup>-1</sup> fresh weight)           | 2005       | —                 | —      | —       | —            |
|                                                      | 2006       | 48.8 $\pm$ 15.9   | 21.3   | 108.1   | Nonnormal    |
|                                                      | 2009       | 110.4 $\pm$ 38.18 | 34.9   | 210.6   | Normal       |
| GSSG SK (nmol g <sup>-1</sup> fresh weight)          | 2005       | —                 | —      | —       | —            |
|                                                      | 2006       | 13.3 $\pm$ 6.2    | 0.0    | 33.2    | Normal       |
|                                                      | 2009       | 36.7 $\pm$ 22.4   | 1.0    | 113.6   | Nonnormal    |
| Percentage GSSG SK                                   | 2005       | —                 | —      | —       | —            |
|                                                      | 2006       | 21.4 $\pm$ 7.5    | 0.0    | 41.4    | Normal       |
|                                                      | 2009       | 25.0 $\pm$ 13.3   | 0.7    | 67.6    | Nonnormal    |
| totGSH SK (nmol g <sup>-1</sup> fresh weight)        | 2005       | —                 | —      | —       | —            |
|                                                      | 2006       | 62.2 $\pm$ 19.4   | 20.4   | 138.9   | Nonnormal    |
|                                                      | 2009       | 147.3 $\pm$ 42.6  | 48.9   | 276.1   | Normal       |
| TAA FL ( $\mu$ mol AsA g <sup>-1</sup> fresh weight) | 2005       | —                 | —      | —       | —            |
|                                                      | 2006       | 3.6 $\pm$ 1.4     | 0.8    | 8.4     | Normal       |
|                                                      | 2009       | —                 | —      | —       | —            |
| TAA SK ( $\mu$ mol AsA g <sup>-1</sup> fresh weight) | 2005       | —                 | —      | —       | —            |
|                                                      | 2006       | 11.5 $\pm$ 4.5    | 1.9    | 25.2    | Normal       |
|                                                      | 2009       | —                 | —      | —       | —            |
| g fresh weight                                       | 2005       | 150.6 $\pm$ 39.5  | 54.5   | 265.7   | Normal       |
|                                                      | 2006       | 113.7 $\pm$ 33.6  | 19.0   | 198.6   | Normal       |
|                                                      | 2009       | 156.6 $\pm$ 47.2  | 43.9   | 291.4   | Normal       |

time taken for cut surfaces to brown (Br\_time) and the final color of the affected tissue (Br\_color). Br\_time ranged from 15 to 265 min and was negatively correlated with the percentage DHA values of flesh tissues (Supplemental Table S5) but not with the flesh totAsA values. Br\_color was positively correlated with the flesh percentage DHA. A strong correlation ( $P < 0.0001$ ) was also found between flesh AsA-totAsA concentrations and firmness and to a lesser extent with fruit percentage dry weight and soluble solids content (measured as °Brix [ $P < 0.01$ ]; Supplemental Table S5).

### QTL Mapping

We identified a total of 27 fruit AsA- and GSH-QTL that were stable for at least two of the three measurement years. These were clustered into four main genomic regions on linkage groups (LGs) 10, 11, 16, and 17 (Fig. 2). The total population variability explained by these AsA-QTL (QTL for AsA, totAsA, and DHA) ranged from 10.7% to 59.5% per trait but was much lower for GSH-QTL (QTL for GSH, totGSH, and GSSG), where population variability ranged between 9.5% and 13.1% per trait (Supplemental Table S6).

#### *QTL for Fruit and Leaf AsA Concentrations*

Twenty flesh AsA-QTL were identified in the cv Telamon parent and 17 in cv Braeburn. These QTL individually explained from 7.6% to 30% of the total population variance (Supplemental Table S6). Clusters of stable (present in two or more years) and significant (log of the odds [LOD] > 3.5) AsA-QTL were identified at the bottom of LG 11 and in the middle of LG 10 in both parental maps (Table II). Additional AsA-QTL on LGs 3, 6, and 16 were not stable across all years, while parent-specific QTL were located on LGs 3 and 17. In the majority of cases, QTL for totAsA collocated with QTL for flesh AsA concentrations in both parents (Fig. 2; Supplemental Table S6). Of particular interest was the QTL cluster for flesh AsA located on LG 11, as these were stable over all three years and explained a relatively high proportion (up to 27.5%) of the total population variability.

Stable clusters of skin AsA-QTL were detected on LGs 9, 10, and 16, which individually accounted for up to 17.3% of population variability, while additional, non-year-stable QTL were found on LGs 2, 3, 11, and 17 (Fig. 2; Supplemental Table S6). The skin AsA-QTL on LGs 11, 16, and 17 collocated with AsA-QTL of flesh, with only the QTL cluster on LG 9 specific for fruit skin AsA-totAsA concentrations.

QTL analysis of leaf AsA, totAsA, and DHA concentrations (2009 only) identified two QTL clusters on LG 6 and 15 (Table III), but only the QTL on LG 6 collocated with that of a fruit AsA-QTL derived from the Telamon parent.

#### *QTL for Fruit and Leaf GSH Concentrations*

From measurements in 2006 and 2009, we identified nine skin and nine flesh GSH-QTL, distributed across

nine LGs (Fig. 2; Supplemental Table S6). LOD scores for these QTL ranged from 3.0 to 3.4 and were generally lower than those observed for AsA-QTL. Only one region in the middle of LG 16 contained QTL that were stable across parents and tissues (fruit flesh, fruit skin, and leaf). In 2009, leaf GSH concentrations were measured, leading to the identification of QTL on LGs 6, 12, and 16 (Fig. 2; Table III), with the QTL for leaf GSSG on LG 16 colocating with fruit AsA-QTL in both parents (Fig. 2).

#### *QTL for Fruit TAA*

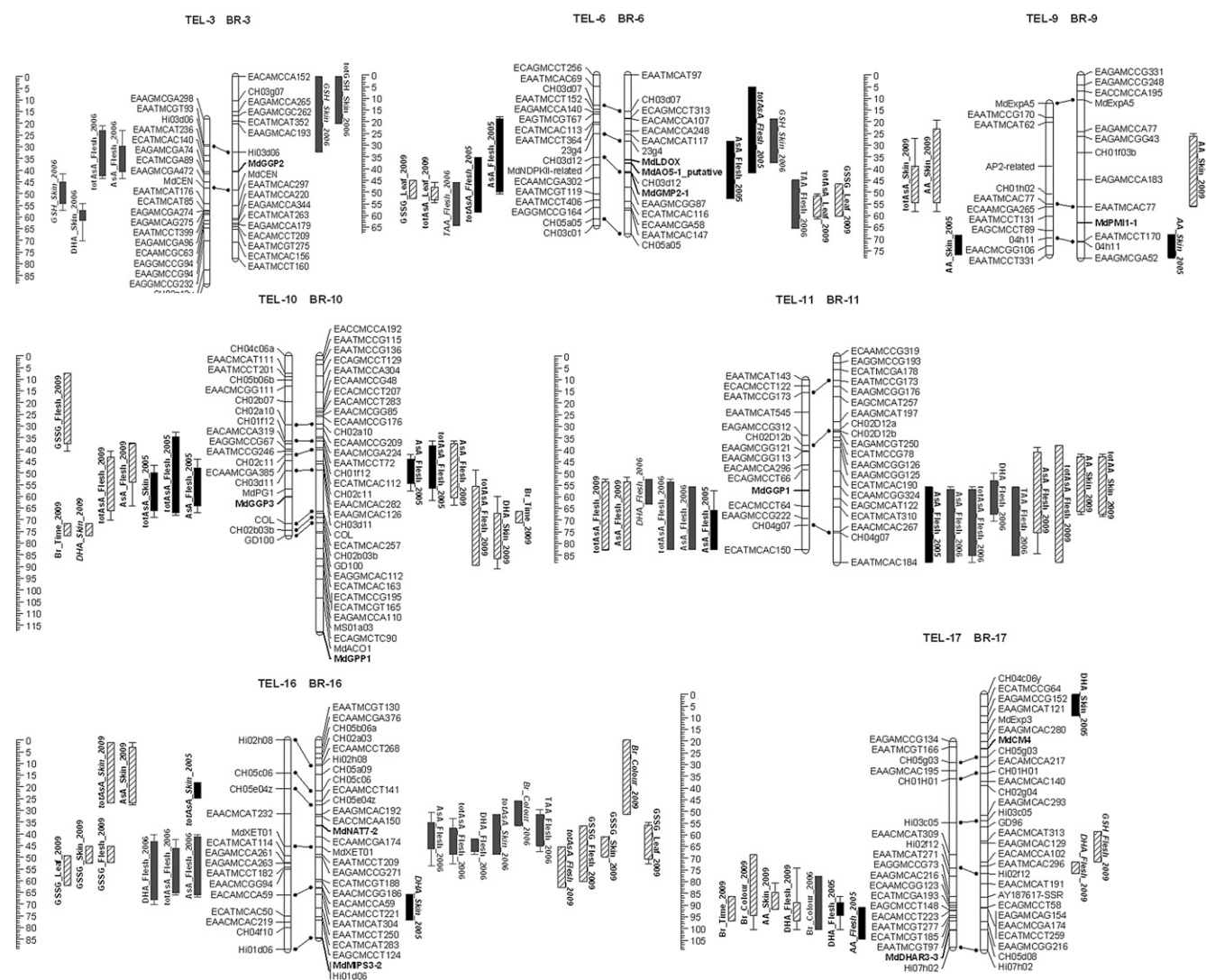
In cv Telamon, two significant TAA-QTL for flesh were detected on LGs 6 and 12, accounting for a total of 23.1% of the population variability (Supplemental Table S6). In cv Braeburn, a major TAA-QTL on LG 11 (LOD = 3.9; population variability of 20.6%), and two minor TAA-QTL on LGs 6 and 16 were identified in flesh tissues, all partially overlapping with QTL for flesh AsA-totAsA.

#### *QTL for Flesh Browning and Fruit Quality Traits*

QTL for susceptibility to flesh browning (Br\_time and Br\_color) were located on LGs 3, 10, 16, and 17 (Fig. 2; Supplemental Table S7). Of these, the QTL for Br\_color on LG 16 of cv Braeburn and on LG 17 of cv Telamon were stable over both years of measurements and collocated with AsA-QTL on these LGs (Fig. 2). On LG 10 (both parents) and on LG 17 (cv Telamon), QTL for Br\_time mapped to the same position as QTL for AsA, totAsA, and DHA concentrations of flesh tissues. Interestingly, stable QTL for flesh browning (Br\_time and Br\_color) at the bottom of cv Telamon LG 17 collocated with the stable QTL for flesh DHA concentrations, but no comparable QTL were detected in the “high-fruit-quality” cv Braeburn parent. A large cluster of fruit AsA-QTL in the middle of LG 10 in both parents collocated with a range of previously identified stable QTL for fruit traits, including fresh weight, flesh soluble solids content, and firmness, in this population (Supplemental Table S7; Kenis et al., 2008).

### Mapping of Candidate Gene Orthologs and Fruit QTL Clusters

From the literature, we identified 22 structural genes involved in AsA biosynthesis, recycling, degradation, and transport. Using the apple reference consensus coding sequence (CDS) set (<http://genomics.research.iasma.it/>), 99 apple orthologs of these genes were identified, based on similarity to Arabidopsis sequences and by phylogenetic analysis using published sequences from several additional plant species (Supplemental Table S8). Following an in silico analysis of their positions in the apple genome (Supplemental Table S8), SNP-based markers (Supplemental Table S9) were developed for the subset of CGs that collocated within our QTL using RNA-Seq data of both parents, allowing these CGs to be mapped onto our cv Tx8 genetic linkage



**Figure 2.** Candidate gene and QTL mapping. Overview is shown for the locations of candidate genes and QTL for mean AsA, totAsA (AsA + DHA), DHA, GSH, totGSH (GSH + GSSG), GSSG, and TAA of fruit flesh and skin tissues as well as leaves and for the rate of flesh browning (Br\_color, Br\_time, Br\_firmness) detected in 2005 (black QTL), 2006 (gray QTL), and 2009 (hatched QTL) on the genetic maps of the apple cultivars Telamon (TEL) and Braeburn (BR). Candidate genes in boldface were mapped in this study and are listed in Supplemental Table S9. Significant QTL (LOD > 3.5) with 1.5-LOD support intervals are shown to the side of each LG, and nonsignificant QTL (LOD < 3.5) are indicated in italics.

maps (Supplemental Fig. S2). A total of 10 AsA CGs were mapped onto the cv Telamon map and 13 onto the Braeburn map. Generally, these CGs mapped to the expected positions predicted from the Golden Delicious genome assembly. The only exception was for *MdDHA3-1* (MDP0000240690; LG 9), where the marker mapped to the homologous region of LG 17, which contains two other paralogous copies of *MdDHA3* (*MdDHA3-2*, MDP0000175246 and *MdDHA3-3*, MDP0000156763).

From the L-Gal biosynthetic pathway, one ortholog of GGP (MDP0000172222, VTC2, *MdGGP1*), with 71% protein sequence identity to Arabidopsis At4g26850.1, mapped to LG 11 of cv Telamon within the cluster of stable flesh AsA-QTL in both parents (Fig. 2). A clone

of this gene from *Malus sylvestris* (GB CN915822) has been shown to have GGP activity when expressed in *Escherichia coli* (Supplemental Table S10). A paralog of *MdGGP1*, *MdGGP2* (MDP0000288088; 69.5% identity to At4g26850.1 and 89.6% identity to *MdGGP1*), mapped to LG 3 of cv Braeburn within the flesh AsA-QTL cluster in cv Telamon that was only detected in 2006. Again, a homologous gene from apple (GB CN939721) has been demonstrated to be active (Supplemental Table S10). A third more distantly related ortholog of Arabidopsis *GPP*, *MdGGP3* (MDP0000191488; 30.1% identity to At4g26850.1 and 34% identity to *MdGGP1*), mapped to LG 10 of cv Telamon and also collocated within a stable AsA-QTL. The markers developed for both *MdGGP1* (2006 and 2009) and *MdGGP3* (2005)

**Table II.** QTL for fruit flesh AsA concentrations

Overview of the stable (detected in at least 2 years) QTL identified for apple fruit AsA, totAsA, and DHA in flesh (FL) tissues in the TxB mapping population. For QTL identified using the Kruskal-Wallis nonparametric mapping functions, we used significance levels as follows: \*\*\* $P = 0.01$ , \*\*\*\* $P = 0.005$ . Gene-specific markers are indicated in italics. An overview of all the QTL for antioxidant concentrations is given in Supplemental Table S6.

| Trait     | cv Telamon |      |      |               |           | cv Braeburn |      |      |             |           |
|-----------|------------|------|------|---------------|-----------|-------------|------|------|-------------|-----------|
|           | LG         | Year | LOD  | Marker        | Variation | LG          | Year | LOD  | Marker      | Variation |
| AsA FL    | 10         | 2005 | 7.03 | <i>MdGGP3</i> | 16.4      | 10          | 2005 | 7.8  | CH03d11     | 23.4      |
|           |            | 2009 | 6.25 | EAATMCCG246   | 13.3      |             | 2009 | 4.5  | CH03d11     | 10.2      |
|           | 11         | 2005 | 4.02 | CH04g07       | 7.6       | 11          | 2005 | 3.9  | CH04g07     | 14.2      |
|           |            | 2006 | 3.5  | CH04g07       | 10.5      |             | 2006 | 4.6  | EAGCMCAT122 | 21.6      |
|           |            | 2009 | 8.5  | <i>MdGGP1</i> | 27.5      |             | 2009 | 5.62 | ECATMCAC190 | 22        |
| totAsA FL | 10         | 2005 | 5.68 | CH03d11       | 15        | 10          | 2005 | 5.98 | CH03d11     | 18.1      |
|           |            | 2009 | 4.5  | CH03d11       | 9.6       |             | 2009 | 4.54 | EAGAMCCA110 | 12.8      |
|           | 11         | 2006 | 4.23 | <i>MdGGP1</i> | 17.5      | 11          | 2006 | 5.72 | EAGCMCAT122 | 24.9      |
|           |            | 2009 | 5.8  | <i>MdGGP1</i> | 18.4      |             | 2009 | 4.08 | ECATMCAC190 | 12.5      |
|           |            | 2009 | **** | EAAGMCAC216   | —         |             | 2009 | ***  | EACAMCCA102 | —         |

were also the best markers describing the respective QTL (Table II). None of the other genes involved in the L-Gal biosynthetic pathway collocated with any of our AsA-QTL, apart from a putative GDP-mannose pyrophosphorylase (GMP or VTC1; MDP0000323050, *MdGMP2-1*) with 83.5% identity to At1g74910, which collocated with the non-year-stable flesh AsA-QTL on LG 6 (Fig. 2).

From the alternative AsA biosynthetic pathways, a marker for a putative D-GalUA reductase (*MdGalUR1*, MDP0000309417; 57.5% identity with At2g37790) mapped to LG 12 of cv Braeburn but did not collocate with any QTL (Fig. 2). However, two *MdGalUR2* sequences (MDP0000135496 and MDP0000857724) mapped in silico within the flesh AsA-QTL on LG 16 (Fig. 2; Supplemental Table S8). It was not possible to develop markers for these others, however, as expression levels were too low to identify unique SNPs from our RNA-Seq data.

We identified six DHAR and six MDHAR apple orthologs involved in the recycling of oxidized DHA (Supplemental Table S8). Of these, *MdDHAR3-3* (65% identity with *AtDHAR3*) mapped to the bottom of LG 17 in cv Telamon and collocated with a stable QTL for flesh DHA concentration (2005 and 2009 data) and a

minor QTL for flesh AsA concentration (2005; Fig. 2). The other paralogs of *MdDHAR3* located in silico to LG 9 (*MdDHAR3-1*, MDP0000240690 and *MdDHAR3-4*, MDP0000530903), also within the QTL intervals for skin AsA concentrations (2005 data). There were no collocations, however, between any of the *MdMDHARs* and antioxidant QTL.

Twenty-one apple orthologs of the NAT protein family were identified in the reference CDS set. Four of these are highly similar to *AtNAT7* and *AtNAT12*, which are the two members considered to be specifically involved in AsA transport, and are located on LGs 11, 13, 15, and 16 (Supplemental Table S8). *MdNAT7-2* (MDP0000320308; 58% identity with *AtNAT7*) mapped to LG 16 of cv Braeburn, collocating with the significant QTL for flesh AsA-totAsA concentrations in both parents (Fig. 2).

Finally, in silico alignment of genes of GSH metabolism (glutathione reductase and glutathione synthase; Supplemental Table S8) identified only one potentially interesting collocation on LG 4 between *MdGS2-2* (MDP0000188669; 61% identity to At5g27380) and GSH-QTL for both flesh and skin tissues detected in cv Braeburn.

**Table III.** QTL mapping for leaf antioxidant concentrations

Overview of QTL identified for apple foliar AsA, totAsA, DHA, GSH, totGSH, and GSSG concentrations in 2009. Borderline-significant QTL ( $2.5 < \text{LOD} < 3.0$ ) are indicated with asterisks.

| Trait  | cv Telamon |    |      |              |           | cv Braeburn |    |      |              |           |
|--------|------------|----|------|--------------|-----------|-------------|----|------|--------------|-----------|
|        | QTL        | LG | LOD  | Marker       | Variation | QTL         | LG | LOD  | Marker       | Variation |
| AsA    | —          | —  | —    | —            | —         | —           | —  | —    | —            | —         |
| totAsA | 1          | 6  | 2.85 | EAATMCGT119* | 8.3       | 1           | 6  | 2.94 | EAATMCAC147* | 7.9       |
| DHA    | 1          | 15 | 3.64 | EAGAMCGC115  | 9.6       | 1           | 15 | 2.84 | ECAAMCCT193* | 7.4       |
| GSH    | 1          | 12 | 4.16 | 28f4         | 10.1      | 1           | 12 | 4.13 | 28f4         | 10.1      |
| totGSH | 1          | 12 | 3.15 | 28f4         | 7.8       | 1           | 12 | 3.34 | EAGMCCG192   | 9.6       |
| GSSG   | 1          | 16 | 3.00 | EAGAMCCA263  | 7.9       | 1           | 6  | 3.37 | CH03d12      | 8.0       |
|        | 2          | 6  | 2.84 | EAATMCGT119* | 7.0       | 2           | 16 | 2.70 | EAGAMCCG271* | 6.4       |

### Sequence Variations in Candidate Regulatory Genes for Fruit AsA Concentrations

High-throughput RNA sequencing (RNA-Seq) was used to detect polymorphisms in the transcript sequences of AsA CGs and to identify allelic SNPs linked to enhanced fruit AsA-totAsA concentrations. Aligning the RNA-Seq reads to the cv Golden Delicious reference consensus CDS set (<http://genomics.research.iasma.it/>) resulted in a mean read coverage for *MdGGP1* (LG 11) of 164.6- and 123.9-fold for cv Braeburn and Telamon, respectively, indicating high transcript expression and good read depth. A total of 10 different SNPs along the 1,704-bp *MdGGP1* cv Golden Delicious reference CDS were identified (Supplemental Fig. S3; Supplemental Table S11). This corresponds to one SNP every 170 bp, which is consistent with previous reports in apple cDNA sequences (one SNP every 149 bp; Chagné et al., 2008). These alignments were used to correct the *MdGGP1* reference gene model, which was found to contain an incorrectly predicted start codon and two extra codons (Supplemental Figs. S3 and S4). As a result, the majority of the SNPs in *MdGGP1* shift into the 5' untranslated region and only two SNPs are found within the CDS, at reference positions 414 and 885 bp. Neither SNP leads to an amino acid change, but SNP<sub>885</sub> is unique to cv Telamon and was developed into a diagnostic SNP marker for mapping and genotyping.

For *MdGGP2* (LG 3), mean read coverage following alignment to the cv Golden Delicious reference consensus CDS set was 122.5- and 111.2-fold for cv Braeburn and Telamon, respectively, again indicating high expression and good read depth. A total of eight SNPs were identified in the *MdGGP2* CDS, of which three were heterozygous in cv Braeburn and six in cv Telamon (Supplemental Fig. S5; Supplemental Table S12). Both cv Telamon and Braeburn possess nonsynonymous SNPs at positions 74 and 169 bp. However, as both SNPs are present in both parents, they are unlikely to be responsible for the parent-specific allelic differences observed. Both also have a homozygous, but different, SNP at reference position 564, which was used for mapping and genotyping.

Mean read coverage of *MdGGP3* following alignment to the cv Golden Delicious CDS was 34.8- and 31.3-fold for cv Braeburn and Telamon, respectively. Both the cv Telamon and Braeburn CDS were noticeably more divergent from the reference CDS than *MdGGP1* and *MdGGP2*, and we identified 79 SNPs in cv Braeburn and 75 SNPs in cv Telamon (Supplemental Fig. S6). A unique nonsynonymous SNP in cv Telamon at reference position 211 was developed for mapping. Both parents contain nonsynonymous SNPs that could influence *MdGGP3* protein function, and both also possess a homozygous 33-bp deletion at reference position 272 at the beginning of intron 4, leading to the loss of 11 amino acids. In addition, in cv Braeburn, a heterozygous single-nucleotide deletion at reference position 986 leads to a frame shift and a different and truncated N-terminal amino acid sequence (Supplemental Fig. S7).

This suggests that cv Braeburn contains an *MdGGP3* allele that codes for a truncated and possibly dysfunctional protein. Alignment of cv Braeburn de novo contigs supports this conclusion, with fragments of both cv Braeburn alleles apparently being present.

The mean read coverage for *MdDCHAR3-1* (LG 9) was 14.0- and 20.0-fold for cv Braeburn and Telamon, respectively; however, these alignments contained a large region of nonspecifically matched reads, so that only a single SNP could be reliably identified. This SNP-based marker mapped to the homologous region on LG 17, which contains both *MdDCHAR3-2* and *MdDCHAR3-3* (MDP0000175246 and MDP0000156763). However, the read coverage for these latter two sequences was too low to allow additional *MdDCHAR3*-specific markers to be developed.

For *MdNAT7-2* (LG 16), the mean read coverage was 33.1- and 34.9-fold for cv Braeburn and Telamon, respectively. However, coverage was not uniform, and it was not possible to generate a full-length consensus sequence in either case. In addition, the consensus read-mapping sequences from both parents showed a considerable degree of polymorphism from the cv Golden Delicious reference CDS, with 252 variants detected in cv Braeburn and 151 in cv Telamon.

### Allele Association Studies

Orthologous CG sequences, which mapped within AsA-QTL, highlighted several genes with a potential role in regulating fruit AsA concentrations. Of particular interest are the paralogous copies of *MdGGP*, as all three were mapped within significant flesh AsA-QTL, and two of these QTL were further stable for at least 2 years. In addition, *MdNAT7-2* colocalized with QTL for flesh antioxidant concentrations on LG 16. To further investigate the involvement of these four genes in the control of apple AsA concentrations, correlations between the allele SNP variants were examined in the TxB progeny as well as in a set of commercial apple cultivars with a wide range of totAsA concentrations.

TotAsA concentrations of flesh tissues in progeny of the TxB population were significantly correlated with SNP variants for *MdGGP1* (all three years), *MdGGP2* (1 year), *MdGGP3* (2 years), and *MdNAT7-2* (2 years; Table IV). Specifically, genotypes homozygous for the T allele (T/T genotypes) of *MdGGP1* at position 885 had statistically significantly higher flesh totAsA concentrations than the T/C genotypes over the three years of measurement. Similarly, G/A genotypes for *MdGGP2* (SNP<sub>564</sub>) or the T/T genotypes for *MdGGP3* (SNP<sub>211</sub>) also had higher flesh totAsA concentration than G/G or T/C, respectively, but only in 1 or 2 years of measurements (Table IV). For *MdNAT7-2* (SNP<sub>92</sub>), the T/G genotypes had also higher fruit totAsA concentrations in 2006 and 2009 compared with G/G genotypes. Comparing the flesh totAsA concentrations of progeny with the "best" and "worst" SNP

**Table IV.** Association between fruit totAsA concentrations and candidate gene allelic variants in the TxB mapping population

Associations are shown between mean fruit flesh totAsA concentrations (nmol g<sup>-1</sup> fresh weight) in progeny of the TxB mapping population and allelic variations in *MdGGP1* (T→C; SNP<sub>885</sub>), *MdGGP2* (G→A; SNP<sub>564</sub>), *MdGGP3* (T→C; SNP<sub>211</sub>), and *MdNAT7-2* (T→G; SNP<sub>92</sub>) across three years of measurements. Statistical significance of the mean fold difference in flesh totAsA concentrations between SNP genotypes was tested with Student's *t* test (\**P* < 0.05, \*\**P* < 0.01, \*\*\**P* < 0.001). Associations were carried out in 71 (2005), 73 (2006), and 85 (2009) individuals of progeny for which both phenotypic and genotypic data were available.

| Gene            | Allelic Variation | cv Telamon | cv Braeburn | Year | Mean Flesh totAsA of SNP Variants in the cv TxB Population |       |     |
|-----------------|-------------------|------------|-------------|------|------------------------------------------------------------|-------|-----|
| <i>MdGGP1</i>   | T→C <sup>a</sup>  | T/C        | T/T         |      | T/C                                                        | T/T   |     |
|                 |                   |            |             | 2005 | 446.4                                                      | 537.9 | *   |
|                 |                   |            |             | 2006 | 352.6                                                      | 429.2 | **  |
| <i>MdGGP2</i>   | G→A               | G/G        | G/A         | 2009 | 561.8                                                      | 786.6 | *** |
|                 |                   |            |             |      | G/G                                                        | G/A   |     |
|                 |                   |            |             | 2005 | 480.7                                                      | 488.1 |     |
| <i>MdGGP3</i>   | T→C               | T/C        | T/T         | 2006 | 357.4                                                      | 416.4 | *   |
|                 |                   |            |             | 2009 | 630.2                                                      | 683.0 |     |
|                 |                   |            |             |      | T/C                                                        | T/T   |     |
| <i>MdNAT7-2</i> | T→G               | G/G        | T/G         | 2005 | 423.5                                                      | 514.9 | *   |
|                 |                   |            |             | 2006 | 340.7                                                      | 425.7 | **  |
|                 |                   |            |             | 2009 | 620.4                                                      | 690.6 |     |
|                 |                   |            |             |      | G/G                                                        | T/G   |     |
|                 |                   |            |             | 2005 | 484.2                                                      | 538.5 |     |
|                 |                   |            |             | 2006 | 355.8                                                      | 419.2 | *   |
|                 |                   |            |             | 2009 | 617.9                                                      | 752.9 | **  |

<sup>a</sup>Genotypes homozygous for the T allele are named as T/T genotypes, while genotypes heterozygous for the T or C allele are named as T/C genotypes.

variant combinations for these genes showed that the best SNP allelic variants resulted in fruit mean totAsA concentrations that were up to 10-fold higher than those carrying the unfavorable allelic variants (data not shown). In all cases, cv Braeburn was the parent that transferred the beneficial allelic variants for high totAsA concentrations.

To validate these associations outside the mapping population, a selection of 22 apple commercial cultivars with known fruit totAsA concentrations (Davey et al., 2004; I. Mellidou and M.W. Davey, unpublished data) were genotyped with the same SNP markers. In the case of *MdGPP2* (LG 3) and *MdNAT7-2* (LG 16), these markers yielded complex melting curve profiles in the high-resolution melting (HRM) SNP assay and so could not be used with this germplasm. This is presumably the result of additional polymorphisms present in the PCR amplicon region. However, the SNP variants in *MdGGP1* and *MdGPP3* were successfully determined for 22 and 11 of these cultivars, respectively (Table V). For the remaining 11 cultivars, the complex melting curve profiles for *MdGPP3* again precluded us from establishing SNP genotypes. Results show that once again the *MdGPP1* T allele and the *MdGPP3* T allele were both associated with higher fruit totAsA concentrations in these cultivars. Furthermore, in the 11 cultivars for which we have the SNP genotypes for both *MdGGP1* and *MdGPP3*, the favorable T+T allele combination was always associated with an average 3-fold higher fruit totAsA concentration and a maximum 6-fold higher concentration than those with the C+C combination.

### Gene Expression Analysis

To understand the basis for the differences in totAsA contents in fruit of cultivars with different SNP alleles, quantitative PCR expression analysis of CGs was carried out in the flesh of fruit from 10 apple genotypes with high or low AsA concentrations (Fig. 3). In all cases, the expression levels of *MdGGP1*, *MdGGP3*, and *MdDHAR3-3* were positively and significantly correlated with fruit totAsA concentrations (Fig. 3, B–G), with the mean differences in their relative expression among the high and low AsA groups being 2.0-, 1.5-, and 2.0-fold, respectively. In contrast, there were no statistically significant differences in the mean relative expression of *MdGGP2* and *MdNAT7-2*, although in both cases, cv Braeburn once again had a higher relative expression than cv Telamon tissues.

Analysis of the RNA-Seq data shows that the mean expression levels of *MdGGP1* (LG 11) and *MdGGP2* (LG 3) are much higher than that of *MdGGP3* (LG 10). Reads per kilobase exon model per million (RPKM) values for *MdGGP1*, *MdGGP2*, and *MdGGP3* in cv Braeburn and Telamon, respectively, were 84.1 and 56.8, 61.4 and 50.4, and 13.9 and 15.3. Although these values are derived from RNA samples isolated during bud and flower development, comparison with the RPKM values of these genes in the low-AsA cv Royal Gala at the mature fruit stage (132 d after full bloom) again shows that *MdGGP1* (RPKM 58) and *MdGGP2* (RPKM 52) are much more strongly expressed than *MdGGP3* (RPKM 5; W.A. Laing, unpublished data). Together with the QTL results, this suggests that

**Table V.** Association between fruit totAsA concentrations and candidate gene allelic variants in commercial cultivars

Associations between totAsA concentrations (nmol g<sup>-1</sup> fresh weight) of the entire (flesh and skin) fruit (Davey et al., 2004) and allelic variations of *MdGPP1* (T→C; SNP<sub>885</sub>) and *MdGPP3* (T→C; SNP<sub>211</sub>) across 22 apple commercial cultivars, as well as across the three TxB individuals with highest or lowest fruit mean totAsA concentrations, are shown. Candidate gene SNP variants were determined by high-resolution melting analysis using the same SNP markers developed for cv Telamon and Braeburn. For *MdGPP3*, the complex melting curve profiles of 11 cultivars prevented SNP genotyping (dashes). Mean fruit totAsA concentrations were determined in 2004 for the commercial cultivars (Davey et al., 2004), while the average measurements of three years (2005, 2006, and 2009) were used for the TxB individuals with the highest and lowest totAsA concentrations.

| No. | Cultivar                              | totAsA | Alleles                             |                                     |                               |
|-----|---------------------------------------|--------|-------------------------------------|-------------------------------------|-------------------------------|
|     |                                       |        | <i>MdGPP1</i> (SNP <sub>885</sub> ) | <i>MdGPP3</i> (SNP <sub>211</sub> ) | <i>MdGPP1</i> + <i>MdGPP3</i> |
| 1   | Ontario                               | 1,448  | T                                   | –                                   | T–                            |
| 2   | Braeburn                              | 1,378  | T                                   | T                                   | T+T                           |
| 3   | TxB 153                               | 1,324  | T                                   | T                                   | T+T                           |
| 4   | Gravenstein                           | 1,148  | T                                   | T                                   | T+T                           |
| 5   | Cox's Orange Pippin                   | 1,132  | T                                   | –                                   | T–                            |
| 6   | TxB 232                               | 1,113  | T                                   | T                                   | T+T                           |
| 7   | Golden Delicious                      | 989.5  | T                                   | T                                   | T+T                           |
| 8   | Greenstar                             | 988.3  | T                                   | C                                   | T+C                           |
| 9   | TxB 283                               | 987.3  | T                                   | T                                   | T+T                           |
| 10  | Kanzi                                 | 889.3  | T                                   | –                                   | T–                            |
| 11  | Angold                                | 857.6  | C                                   | T                                   | C+T                           |
| 12  | Goldrush                              | 815.0  | T                                   | –                                   | T–                            |
| 13  | Jonagold                              | 798.0  | T                                   | C                                   | T+C                           |
| 14  | Idared                                | 767.0  | T                                   | –                                   | T–                            |
| 15  | Florina                               | 741.1  | T                                   | –                                   | T–                            |
| 16  | Liberty                               | 728.8  | C                                   | –                                   | C–                            |
| 17  | Pink Lady                             | 651.1  | C                                   | –                                   | C–                            |
| 18  | Fuji                                  | 628.0  | C                                   | C                                   | C+C                           |
| 19  | Elstar                                | 545.6  | C                                   | C                                   | C+C                           |
| 20  | TxB 300                               | 461.7  | C                                   | C                                   | C+C                           |
| 21  | Alkmene                               | 457.4  | C                                   | –                                   | C–                            |
| 22  | Prima                                 | 443.8  | C                                   | C                                   | C+C                           |
| 23  | Merlijn                               | 442.3  | T                                   | –                                   | T–                            |
| 24  | Gala                                  | 437.7  | C                                   | –                                   | C–                            |
| 25  | TxB 299                               | 401.4  | C                                   | C                                   | C+C                           |
| 26  | Priscilla                             | 400.9  | C                                   | C                                   | C+C                           |
| 27  | TxB 126                               | 387.8  | C                                   | C                                   | C+C                           |
| 28  | Telamon <sup>a</sup>                  | 233.4  | C                                   | C                                   | C+C                           |
|     | Mean totAsA of cultivars carrying T   |        | 997.3                               | 1,114                               |                               |
|     | Mean totAsA of cultivars carrying C   |        | 510.4                               | 528.9                               |                               |
|     | Mean fold difference                  |        | 2                                   | 2                                   |                               |
|     | Maximum fold difference               |        | 6                                   | 6                                   |                               |
|     | Mean totAsA of cultivars carrying T+T |        |                                     |                                     | 1,157                         |
|     | Mean totAsA of cultivars carrying C+C |        |                                     |                                     | 437.8                         |
|     | Mean fold difference                  |        |                                     |                                     | 3                             |
|     | Maximum fold difference               |        |                                     |                                     | 6                             |

<sup>a</sup>totAsA concentration of cv Telamon measured in 2006.

*MdGPP1* expression is particularly important for the regulation of tissue flesh AsA concentrations.

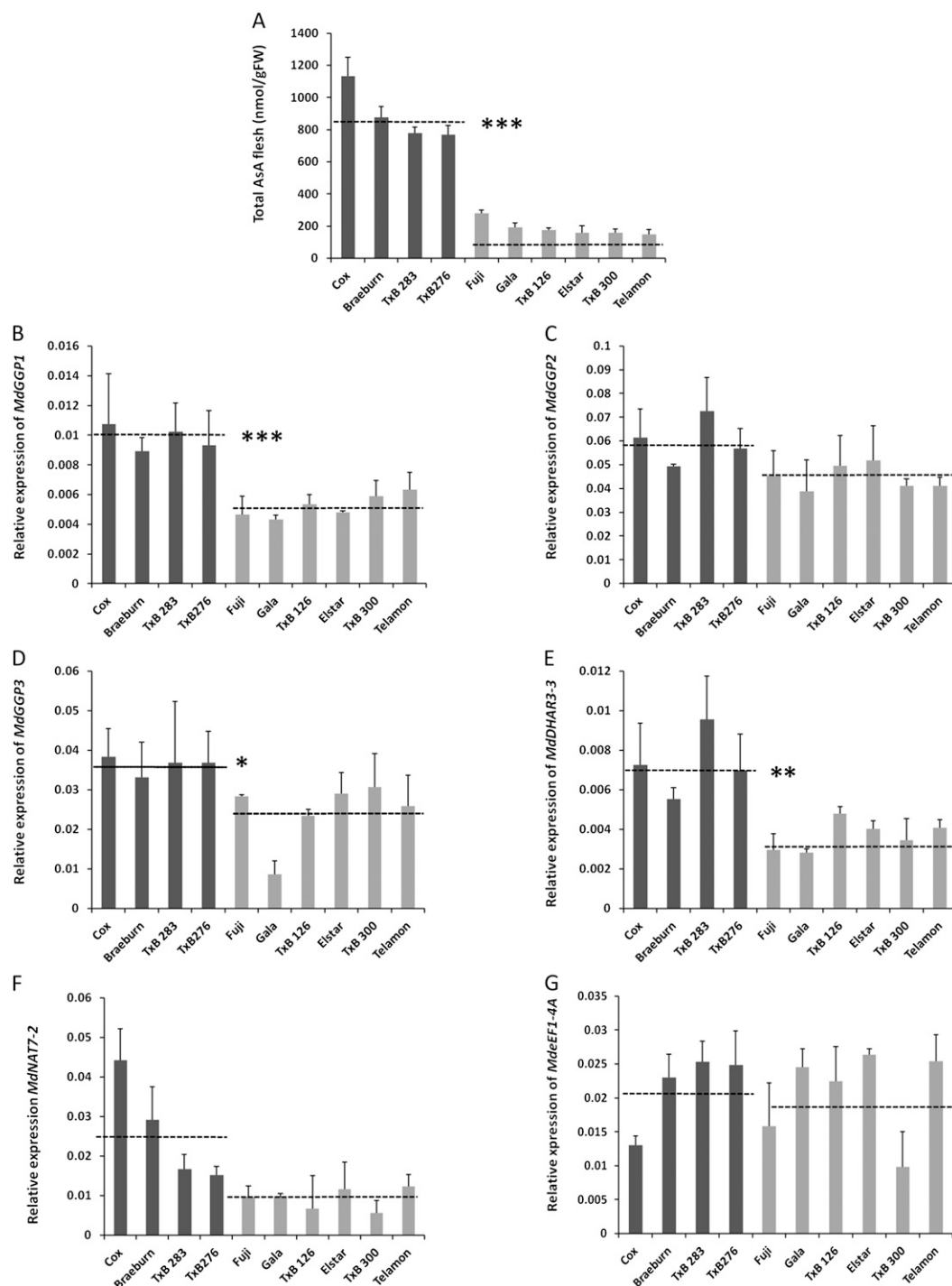
## DISCUSSION

### Antioxidant Concentrations and QTL in the cv TxB Mapping Population: Genetics versus Environment

The absolute concentration of antioxidants varied considerably from year to year, with totAsA and totGSH concentrations being greatest in 2009. This is presumably due to the higher light and temperatures experienced prior to harvest in 2009, factors that are known to influence totAsA concentrations (Davey et al., 2000,

2007; Dumas et al., 2003). However, there was no clear correlation between totAsA concentrations and mean daily temperatures or sunlight hours across the years, underlying the fact that it is difficult to dissociate the individual contributions of temperature, light, development, and genetics from the regulation of antioxidant pools. Nonetheless, the fact that stable (present in at least two measurement years) fruit flesh AsA-QTL (LGs 10, 11, and 17) were found demonstrates a strong genetic component to the control of fruit AsA concentrations in apple.

Environmental factors such as light intensity are expected to have a larger impact on AsA concentrations in the skin compared with flesh tissues and



**Figure 3.** Gene expression studies. Changes in flesh totAsA concentrations (nmol g<sup>-1</sup> fresh weight; A) and expression levels of *MdGGP1* (B), *MdGGP2* (C), *MdGGP3* (D), *MdDHA3-3* (E), *MdNAT7-2* (F), and *MdEIF1-4A* (G) in the mature fruit of apple genotypes with high totAsA concentrations (dark gray bars) and low totAsA concentrations (light gray bars). Quantitative PCR values were normalized against *18S-rRNA* and *Actin*. The error bars represent SD of three biological replications. Dashed lines represent mean flesh totAsA concentrations or mean relative gene expression of the high- and low-AsA groups of genotypes. The significance of the mean relative gene expression of each group was tested with Student's *t* test in SAS (\**P* < 0.05, \*\*\**P* < 0.001).

therefore could mask the underlying genetic control mechanisms. Despite this, significant QTL for skin AsA and totAsA concentrations were identified, and most of these collocated with flesh AsA-QTL. Only the QTL cluster on LG 9 (both parents) and on the cv Telamon LG 16 were specific for skin AsA concentrations. These regions on LG 9 and LG 16 have both been shown to contain major QTL for apple fruit polyphenolic contents (Chagné et al., 2007, 2012; Khan et al., 2012). We have previously reported that the sun-exposed (red) side of apples has higher anthocyanin and AsA concentrations and is more resistant to both biotic and abiotic stress than the shaded (green/yellow) side (Davey et al., 2004, 2007), and recently, Bulley et al. (2012) demonstrated that transgenic fruits from tomato and strawberry with elevated AsA contents also had an approximately 50% larger polyphenolic pool. Finally, we report here positive correlations between TAA, AsA, and totAsA concentrations in both flesh and skin tissues and collocations between flesh AsA- and TAA-QTL on LGs 6, 11, and 16. Therefore, the skin-specific AsA-QTL on LGs 9 and 16 may be related to interaction with the polyphenol contents in apple skin tissues. In this regard, the recent report indicating that the accumulation of anthocyanins in *Arabidopsis* leaves is fine-tuned by the AsA redox state is of particular interest (Page et al., 2012).

For the first time, to our knowledge, we also present QTL for fruit GSH concentrations, but apart from QTL for flesh GSSG concentrations on LG 16 that collocated with AsA-QTL in both parents, all the other QTL had low LOD scores, and no significant associations between AsA- and GSH-QTL were detected.

The active transport of foliar AsA to developing sink tissues has been demonstrated previously (Franceschi and Tarlyn, 2002; Hancock et al., 2003; Tedone et al., 2004). While both apple fruit and leaf tissues have been shown to be capable of AsA biosynthesis (Davey et al., 2004; Razavi et al., 2005; Li et al., 2010), there were no significant correlations between fruit and leaf totAsA concentrations and only one collocation between leaf and fruit AsA-QTL on LG 6. This would also need to be confirmed, as it is based on only 1 year of measurements. Regardless, these results indicate that screening progeny on the basis of foliar AsA concentrations to identify cultivars rich in fruit vitamin C concentrations will not be highly informative.

#### Links between Vitamin C Concentrations, Flesh Browning, and Other Fruit Quality Traits

Susceptibility to browning has been linked to flesh DHA concentrations in apple (Davey et al., 2006), and in pears, the development of internal postharvest browning disorder occurs when AsA concentrations decline below a certain threshold value (Franck et al., 2003). Here, the time taken for cut apple flesh to complete browning and the final color intensity following browning were both correlated with DHA concentrations and the percentage DHA of flesh tissues. In addition, stable QTL for

browning collocated with stable QTL for flesh DHA concentrations on LG 17. This browning QTL was only detected in cv Telamon, which produces poor-quality fruit and in which the flesh percentage DHA was on average 3-fold higher than in cv Braeburn (data not shown). Taken together, these results suggest that factors that lead to an oxidized AsA pool are associated with susceptibility to flesh browning.

In cherry tomatoes, fruit firmness and AsA concentrations have been reported to be closely correlated (Gilbert et al., 2009), and here, we show that firm apples also generally contain higher flesh AsA concentrations. Flesh AsA-totAsA concentrations and percentage DHA are also significantly correlated with fruit flesh soluble solids content, suggesting a link to the supply of sugar substrates for AsA biosynthesis. However, the significance of this is not clear, as a large cluster of QTL for fruit physiological traits, including fruit firmness, soluble solids content, fresh weight, and flesh browning, all collocated with the fruit flesh AsA-QTL on LG 10; therefore, the correlation with fruit AsA-totAsA concentrations could be due to indirect effects on fruit physiology and quality.

#### Candidate Genes Regulating AsA Concentrations

##### *Candidate Regulatory Genes from the AsA Biosynthetic Pathways*

GMP (VTC1) catalyzes the conversion of D-Man-1-P to GDP-D-Man (Fig. 1), and we observed a collocation between a putative *MdGMP* (*MdGMP2-1*) and the flesh AsA-QTL cluster on LG 6. However, these QTL explained only a relatively low proportion of the population variability (10%) and were detected only in 1 year. In tomato (Ioannidi et al., 2009) and kiwifruit (Bulley et al., 2009), fruit AsA concentrations were also not correlated with *GMP* expression.

GME has been proposed to be critical for the regulation of plant AsA concentrations by several groups (Wolucka and Van Montagu, 2007; Gilbert et al., 2009), and in tomato, *SIGME1* collocated with a cluster of stable fruit AsA-QTL (Stevens et al., 2007), while in apple, *MdGME* transcript levels are highly correlated with AsA concentrations during apple fruit development (Li et al., 2011). However, the data presented here support the results obtained in peach (*Prunus persica*; Imai et al., 2009) and tomato (Ioannidi et al., 2009), which suggest that this step is not a major point of control in mature apple fruit, at least under our field conditions.

GGP (VTC2; Fig. 1) catalyzes the first committed step of AsA biosynthesis and has also been suggested to be the rate-limiting step for AsA biosynthesis in plants (Linster and Clarke, 2008; Bulley et al., 2009, 2012). Our results here provide strong evidence that *MdGGP1* is the only structural gene of AsA metabolism that is tightly linked to flesh AsA concentrations in apple and that this is independent of the environmental conditions, since significant AsA-QTL were detected over all three years of measurement. While we cannot exclude

the possibility that these QTL regions contain other regulatory genes, it is interesting that the two other *MdGGP* paralogs (*MdGGP2* on LG 3 and *MdGGP3* on LG 10) also mapped within fruit AsA-QTL.

The next step of the L-Gal pathway is catalyzed by *GPP* (VTC4). Although *GPP* has been suggested to be important in regulating AsA concentrations in tomato fruit (Ioannidi et al., 2009) and apples (Li et al., 2011), these conclusions are not supported by our results. None of the other genes involved in the main AsA biosynthetic pathway (*PMI*, *PMM*, *GalDH*, or *GLDH*) mapped to positions located within our AsA-QTL, suggesting that these have no major control over AsA homeostasis in mature apple fruit under Belgian field conditions.

From the alternative (uronic acid) AsA biosynthetic routes (Fig. 1), the reaction catalyzed by *GalUR* (Enzyme Commission [EC] 1.1.1.203) in strawberry has been the most extensively studied (Agius et al., 2003). D-Galacturonate has been suggested to be an alternative substrate for the synthesis of AsA in fruit skin by precursor feeding experiments in apple fruit tissues (Li et al., 2008), and biosynthesis proceeding via *GalUR* has been suggested to play a more significant role in the later stages of fruit ripening when breakdown of cell wall pectin takes place (Melino et al., 2009; Cruz-Rus et al., 2011; Badejo et al., 2012). The in silico colocation of two *MdGalUR* paralogs within flesh AsA-QTL clusters on LG 16 of both parents supports a possible role for *GalUR* in mature fruits, but there were no other colocations between genes of the alternative AsA biosynthetic pathways and AsA-QTL.

To summarize this section, except for paralogs of the first dedicated step of the L-Gal biosynthetic pathway (*MdGGP1-3*), our results do not support a key role for any of the other genes of the main L-Gal biosynthetic pathway. However, as reactions before the *GPP* step also supply substrates for cell wall polysaccharide biosynthesis, mutations in these genes could affect the flux of substrates through to *GPP*.

#### *CGs from the AsA Recycling Pathway*

DHA, the oxidized form of AsA, is unstable and can be readily degraded if not rereduced by the activity of MDHAR (EC 1.6.5.4) and/or DHAR (EC 1.8.5.1). In tomato, an ortholog of MDHAR (*SIMDHAR3*) has been found to colocate with a stable QTL for fruit AsA contents (Stevens et al., 2007), and MDHAR enzyme activity has been associated with elevated AsA concentrations under chilling stress (Stevens et al., 2008). In apple, we found no colocation between *MdMDHARs* and any antioxidant QTL. However, *MdDHAR3-3* mapped to the bottom of cv Telamon LG 17 within a highly significant, stable QTL for flesh DHA concentrations, and *MdDHAR3-1* mapped in silico within QTL for skin AsA concentrations in LG 9, suggesting that orthologs of DHAR could be important in helping to maintain the reduced AsA pool in apple fruit. The colocation of *MdDHAR3-3* within a stable QTL for flesh browning

suggests that genetic regulation of redox status of the AsA pool via DHAR is important for postharvest fruit quality traits in apple.

#### *CGs Related to Intercellular AsA Transport*

Intercellular AsA transport via the NATs could represent an important mechanism of maintaining sink tissue AsA concentrations (Maurino et al., 2006). *MdNAT7-2* mapped to the same position as significant fruit AsA-QTL on LG 16 of the cv Braeburn genetic map, and another ortholog of NAT (*MdNAT12-2*) collocated in silico with a QTL for leaf DHA concentrations on LG 15 of both maps. As NATs are highly expressed in vascular tissues (Maurino et al., 2006) and they collocated with AsA-QTL in both fruit and leaf tissues, these results indicate that they may be involved in the long-distance transport of AsA in apples.

#### **Molecular Markers for Fruit AsA Concentrations**

Here, we show that allelic variations in the nucleotide sequences of *MdGGP1*, *MdGGP3*, and *MdNAT7-2* are associated with up to 10-fold differences in totAsA concentrations of fruit from individuals of our cv TxB mapping population. The remaining *MdGGP* paralog, *MdGGP2*, also mapped within an AsA-QTL cluster on LG 3, but this cluster was only detected in 2006. Importantly, fruit totAsA concentrations of 22 commercial apple cultivars were also associated with the same allelic variations in *MdGGP1* and *MdGGP3*, and the T+T combination of alleles for *MdGGP1* and *MdGGP3* is consistently associated with an up to 6-fold higher fruit totAsA concentration. Sequence analysis of the (corrected) *MdGGP1* CDS showed that there are no non-synonymous SNPs (Supplemental Table S11), while in *MdGGP2*, nonsynonymous SNPs result in three amino acid changes (Supplemental Table S12), but none of these are predicted to alter structure or function. By comparison, *MdGGP3*, *MdDHAR3-3*, and *MdNAT7-2* CDS contain multiple synonymous and nonsynonymous SNPs. Further study of the impact of these sequence differences, however, was limited by the complexity and relatively low coverage of the *MdDHAR3-3* and *MdNAT7-2* transcript sequences. Since the SNPs in *MdGGP1* and *MdGGP2* are not associated with altered protein function, it is likely that the observed allelic associations are due to linkage with polymorphisms in the promoter region that alter allele expression. Gene expression analyses in a range of phenotypically diverse cultivars demonstrated that *MdGGP1* and *MdGGP3* as well as *MdDHAR3-3* expression levels are all strongly correlated with fruit totAsA concentrations (Fig. 3). Expression levels of *MdGGP2* (LG 3) or *MdNAT7-2* (LG 16), in contrast, were not significantly correlated, but this could be because they mapped within a non-year-stable AsA-QTL cluster and the expression analyses were carried out in fruit harvested in a different year. Conceivably, therefore, these CGs might still be important only under certain environmental/developmental conditions. Finally,

our RNA-Seq data demonstrate that mean expression levels of the third paralog of *MdGGP* (*MdGGP3*) are considerably lower than the expression of the other two gene copies in developing tissues as well as in mature fruit. This, in combination with the high number of SNPs and other polymorphisms, suggests that *MdGGP3* is less important for the regulation of tissue AsA concentrations and that the translation product may even be nonfunctional.

## CONCLUSION

High fruit AsA concentration is considered to be a desirable trait for consumers, but in apple, as in other hard fruit species, the long juvenile period of plants means that fruit-specific traits cannot be determined until progeny reach maturity around 3 to 5 years after germination. As such, there is much interest in the development of genetic markers for the early selection and/or screening of progeny for fruit AsA concentrations. Our QTL, CG mapping, and analysis of allelic variations across a selection of germplasm show that *MdGGP* paralogs are consistently linked to fruit totAsA concentrations, clearly indicating a major role for these genes. Gene expression analyses suggest that the regulation of flesh AsA concentrations in apple is primarily through control of the expression levels of *MdGGP1*, *MdGGP3*, as well as *MdDHAR3-3*. The SNP-based markers developed here are thus excellent candidates for the early screening of progeny for increased fruit AsA concentrations. Despite conflicting reports in the literature, our work does not support a major role for the other biosynthetic genes in apple fruit AsA homeostasis, at least under Belgian climatic conditions. However, we propose that *MdDHAR3-3* helps regulate the redox state of the AsA pool in apple fruit and that increased flesh DHA concentrations are associated with susceptibility to flesh browning.

## MATERIALS AND METHODS

### Mapping Population

An F1 mapping population consisting of 257 individuals was created from a cross between the apple (*Malus × domestica*) cultivars Telamon and Braeburn, using cv Telamon as the female parent (Kenis and Keulemans, 2005). For the 2005 and 2006 work, a population of mature trees more than 4 years old was used. Trees were grown on M.9 rootstock at the Rillaar Experimental Field Station in Aarschot, Belgium. For the 2009 work, a regenerated copy of the same population was propagated on M.27 rootstock, grown on the same field.

### Harvest and Meteorological Measurements

Fruit from individual progeny were harvested up to twice per week when considered commercially ripe. Ripeness was assessed by experts based on ripening parameters. A minimum of 10 healthy apples per genotype were sampled for AsA, totAsA, GSH, totGSH, and TAA analysis and processed essentially as described previously (Davey et al., 2006). Fruit were immediately transported to the laboratory for analysis. Leaf tissue from 180 genotypes was collected before fruit harvest and stored at  $-80^{\circ}\text{C}$  until further analysis. Mature apple fruit were also collected in 2011, pooled into three biological replications, each one consisting of at least three healthy apples, and processed immediately for AsA measurement or stored at  $-80^{\circ}\text{C}$  for gene expression studies. Data for mean daytime air temperature ( $^{\circ}\text{C}$ ) and sun radiation ( $\text{W m}^{-2}$ ) during a period of 5

weeks before the harvest period, starting on August 1, were obtained from the university meteorological station (Heverlee), located 25 km from the experimental field.

### Fruit Physiological Measurements and Flesh Browning

For all years of QTL analysis, fruit fresh and dry weight were measured immediately after harvest. Flesh firmness was determined using a penetrometer fitted with an 11-mm-diameter plug, while at the same time, the juice released by penetration of the probe was used to measure the soluble solids concentration (mainly sugars; °Brix) using a digital refractometer (Pocket PAL-1; Atago), essentially as described by Kenis et al. (2008). In 2006 and 2009, the susceptibility of the apple flesh to browning was assessed as Br\_time, and Br\_color was also evaluated (Davey and Keulemans, 2009).

### AsA and GSH Analysis

Samples for metabolite analysis were obtained from skin and flesh fractions from 10 individual fruit by blending using a metaphosphoric acid/EDTA/polyvinylpyrrolidone extraction, essentially as described previously (Davey et al., 2003, 2006). Extracts were filtered and injected into an HPLC system (Waters 2690 separation module) equipped with a  $\text{C}_{18}$  rocket column (Grace; 53 mm  $\times$  7 mm). AsA and GSH were detected at 243 and 197 nm, respectively (Waters 996 photodiode array detector). AsA analysis usually occurred on the same day within a few hours of harvest, although at peak harvest time it was sometimes necessary to store apples for a maximum of up to 48 h in a cool cell at  $2^{\circ}\text{C}$  before extraction. In total, the fruit from 136, 140, or 163 individual progeny were analyzed for AsA, totAsA, and DHA concentrations in the season of 2005, 2006, and 2009, respectively, whereas GSH, totGSH, and GSSG measurements of the same samples were carried out in fruit only in 2006 and 2009.

### TAA

The acid-soluble TAA of the fruit tissues harvested in 2006 was determined as the sum of the 2,2-azinobis (3-ethylbenzthiazoline-6-sulfonic acid)-reactive substances present in aliquots of the same extracts used to measure AsA and GSH concentrations (Davey et al., 2007).  $A_{734}$  was measured in three technical replicates of all samples using the Multiskan Spectrum Microplate spectrophotometer (Thermo Labsystems). AsA was used as an external standard, and the TAA of each extract was expressed as AsA equivalents.

### Linkage Map Analysis and Development of a Bin Mapping Set

The available genetic maps of cv Telamon and Braeburn (Kenis et al., 2008) were further saturated with 14 additional microsatellite loci and CG-based markers to improve marker density in key chromosome regions. The updated linkage maps of cv Telamon and Braeburn were constructed using the JoinMap 3.0 software (Van Ooijen and Voorrips, 2001), essentially as described previously (Davey et al., 2006; Kenis et al., 2008).

A total of 324 and 339 molecular markers were initially ordered in 17 linkage groups, corresponding to the 17 apple chromosomes. To improve the resolution of QTL analysis, markers that were less informative (scored on less than 50% of genotypes), markers with distorted segregation, or markers with two alleles heterozygous in both parents (*hik × hik* segregation) were discarded, unless the gap between neighbor markers was greater than 10 centimorgans. A subset of 298 markers was used to construct the cv Telamon genetic map, with 96% of the markers highly informative (*ab × aa*, *ef × eg*, *ab × cd*), whereas 313 markers were employed for the cv Braeburn genetic map, which contained less than 10% less informative markers (*hik × hik*). A bin set of 14 highly informative genotypes was created essentially as described by Chagné et al. (2008). Successful markers were considered those that were heterozygous in at least one parent, had a normal segregation ratio in the bin set, and mapped onto the expected position on the bin maps.

### QTL Analysis

QTL for AsA, GSH, and fruit physiological traits were identified using phenotypic data from 2005, 2006, and 2009 and updated genetic maps of the TxB population. QTL mapping was carried out using the MapQTL 4.0

software (Van Ooijen et al., 2002). QTL were detected using interval mapping in combination with restricted multiple QTL model mapping (rMQM) for normally distributed traits with a step size of 1 centimorgan or using the Kruskal-Wallis nonparametric test for nonnormally distributed ones. In rMQM, the most informative marker with the highest LOD score within the intervals of a QTL was selected as a cofactor, and multiple rMQM runs were performed until the cofactors selected were stable, as described previously (Davey et al., 2006). The LOD threshold for the significance of a QTL was selected based on the genome-wide level, using 1,000 permutations at 90% for each trait (around 3.5 or 3.0 for fruit and leaf tissues, respectively). A QTL was classified as significant/nonsignificant and major/minor based on a LOD threshold of 3.5 and 20% of explained population variability. QTL identified were described by the marker with the highest LOD score in the corresponding QTL region, and QTL regions were defined as the 1.5-LOD support interval. Linkage maps and QTL were drawn using MapChart 3.0 software (Voorrips, 2002).

## Candidate Gene Mapping

The sequences of AsA-related CGs from Arabidopsis or other plant species were retrieved from GenBank (<http://www.ncbi.nlm.nih.gov/genbank/>), PLAZA 2.0 (<http://bioinformatics.psb.ugent.be/plaza/>), Phytozome Genome Browser (<http://www.phytozome.net>), or the SOL Genomics Network (<http://solgenomics.net>). Orthologous sequences were retrieved from the cv Golden Delicious assembly (Velasco et al., 2010) using BLASTX versus the apple consensus CDS set. Gene positions on the genome assembly and their sequences were extracted from the IASMA genome browser (<http://genomics.research.iasma.it>). The amino acid sequences for each CG were aligned using the BLOSUM62 matrix, and phylogenetic analysis was conducted using the neighbor-joining method and 1,000 bootstrap replicates in Geneious software version 5.4.6 (Drummond et al., 2011) in order to determine the number of copies present in apple for each Arabidopsis accession. The positions of these homologs were aligned in silico across the 17 linkage groups of the cv TxB map, based on the physical positions of framework simple sequence repeats.

## SNP Marker Development

RNA-Seq data from cv Telamon and Braeburn were used to identify SNPs and polymorphisms in CG sequences. In total, 48.1 and 73.6 million 75-bp paired end reads, respectively, were generated from RNA pooled from eight different developmental stages of tree bud development and two tissue types. Reads were trimmed and then mapped against the apple consensus CDS set, downloaded from the GDR Web site (<http://www.rosaceae.org/>), using default mapping parameters within CLCBio Genomics Workbench version 5.5. CLCBio software was also used to carry out the de novo assembly of reads into contigs for each cultivar separately, again using default alignment parameters and the automatic scaffolding function. SNPs were called relative to the cv Golden Delicious reference CDS on the basis of using a minimum read depth of 6-fold and a percentage proportion coverage of greater than 25%. Finally, to control gene annotation, the consensus sequences from both the mapped reads and the de novo contigs were extracted and aligned against the annotated genomic DNA contigs used to create the apple genome assembly, again downloaded from GDR and using the Geneious version 5.6 software package. Functional descriptions of consensus CDS and translated protein sequences were confirmed using Blast2GO software (<http://www.blast2go.com/b2ghome>) as well InterPro (Geneious version 5.6). To validate gene function, clones of GGPs were expressed in *Escherichia coli*, and GGP activity was measured essentially as described previously (Laing et al., 2007).

PCR primers were developed to identify length polymorphisms, or flanking SNPs, to allow progeny genotyping using the HRM technique (Chagné et al., 2008; Studer et al., 2009). Gene-specific SNP primer pairs were designed using Primer3 software version 4.0 (<http://frodo.wi.mit.edu/primer3/>) employing the following conditions: product size between 75 and 200 bp, with optimum 120 bp spanning the identified SNPs; GC% ranging from 40% to 55%; self-complementarity and 3' self-complementarity were set at 4 and 1, respectively; and all the other parameters were set at default. All CG-derived markers developed were screened in the cv TxB bin mapping set and both parents.

Genomic DNA from leaves was extracted from approximately 100-mg powdered aliquots using the Genomic DNA Purification kit (MBI Fermentas). HRM conditions were as follows: the PCR for each sample was carried out in 15  $\mu$ L containing the HRM Master Mix (Type-it HRM PCR Kit; Qiagen), 10 to 20  $\mu$ M of each primer, and 5 ng of genomic DNA. Cycling conditions were incubation at 95°C for 5 min, followed by 40 cycles of 95°C (10 s) and 55°C (30

s). HRM analysis was followed immediately by denaturation from 65°C to 95°C, rising by 0.1°C in each step. The HRM analysis was carried out in a Rotorgene Q thermocycler software (Qiagen). Markers that mapped to their expected positions and gave an expected segregation rate were further screened over 100 individuals from the cv TxB population to construct new genetic maps, as described previously.

## Allelic Typing and Gene Expression Studies

Gene expression studies were performed following Minimum Information for Publication of Quantitative Real-Time PCR Experiments guidelines (Bustin et al., 2009). Total RNA was isolated from flesh fruit tissues using a modified cetyl-trimethyl-ammonium bromide method (Gasic et al., 2004) and the RNeasy Plus Mini Kit following the manufacturer's instructions (Qiagen). The purity of total RNA extracted was determined as the 260/280-nm or 260/230-nm ratio with NanoDrop 2000 (Thermo Scientific), and the integrity was checked by electrophoresis on a 1% agarose gel (Gel Doc EZ Imager; Bio-Rad). One microgram of total RNA was reverse transcribed to cDNA with SuperScript II (Invitrogen) using random oligo(dT) primers. The expression profiles of CGs considered to be key control points for AsA accumulation (*MdGGP1*, *MdGGP2*, *MdGGP3*, *MdDHA3-3*, and *MdNAT7-2*; primer pair sequences are listed in Supplemental Table S9) were validated by real-time quantitative PCR using SYBR Green I technology on a Rotor Gene Q (Qiagen). *18S-rRNA* (5'-GTTACTTTTAGGACTCCGCC-3' and 5'-TTCCTTTAAGTTTCAGCCTTG-3'), *Malus* spp. *eIF-4A* (5'-ATCAGGCTCATCCGTGT-3' and 5'-AGCAACACCCTTCCTTCC-3'; Zubini et al., 2007), and Arabidopsis *Actin11* (5'-GGACCTTGACGGCCGTGACC-3' and 5'-AACCTCCGGGACGCGGAATC-3') were used as reference genes. The stability of the expression of the references genes was assessed with NormFinder software (<http://www.mdl.dk/publicationsnormfinder.htm>). All reactions were set up in duplicate, containing a 1- $\mu$ L cDNA template (50 ng), 7.5  $\mu$ L of Absolute QPCR SYBR Green Mix, and 1  $\mu$ L of each primer (3.75 mM) in a final volume of 15  $\mu$ L. The cycling conditions were as follows: denaturation step at 95°C for 10 min, followed by 45 cycles of denaturation at 95°C for 20 s, annealing at 63°C for 20 s, and extension at 72°C for 20 s. A melting curve analysis was performed, ranging from 55°C to 95°C, with temperature increasing in steps of 0.5°C s<sup>-1</sup>. The criteria of acceptance for reaction efficiency ranged from 0.83 to 1.03 and both *r* and *r*<sup>2</sup> > 0.97. For each run, a standard curve was included based on cDNA pooled from all samples to be analyzed in a range of different dilutions. The relative quantification of expression levels was performed using the comparative cycle threshold method (Pfaffl, 2001). Two technical replications were performed for each of the three biological replications per sample. All expression data were calculated as an expression ratio relative to the geometric mean of the expression of *18S-rRNA* and *Actin*, as these showed the highest stability in our sample set, with stability values (Vandesompele et al., 2002) of 0.34 and 0.47, respectively.

## Descriptive Statistical Analysis

Statistical analysis of all traits was carried out using the SAS 9.2 software package (SAS Institute). Mean values, SD values, ranges, and skewness of the F1 progeny were calculated for each year and trait, while the normality of the distributions was tested using the Shapiro-Wilk test (Shapiro and Wilk, 1965). Pearson correlation coefficients were employed to compare mean values of the traits over the three years. The significance of the mean fold difference in totAsA concentrations of genotypes carrying different alleles/allele combinations, as well as the significance of the mean relative expression of CGs of high/low AsA genotypes, were tested with Student's *t* test (\**P* < 0.05, \*\**P* < 0.01, and \*\*\**P* < 0.001).

## Supplemental Data

The following materials are available in the online version of this article.

**Supplemental Figure S1.** Climatic conditions prior to fruit harvest.

**Supplemental Figure S2.** Genetic linkage maps of cv Telamon (TEL) and Braeburn (BR) used for QTL analysis.

**Supplemental Figure S3.** Alignment of the MDP0000172222 (*MdGGP1*) consensus CDS from apple cultivars Braeburn (Br), Telamon (Tel), and Golden Delicious (GD) with the full-length EST sequence from cv Royal Gala (gi|225380879).

**Supplemental Figure S4.** Alignment of the MDP0000172222 (*MdGGP1*) consensus CDS from apple cultivars Braeburn (Br), Telamon (Tel), and

the full-length EST sequence from cv Royal Gala (gi|225380879) with the genomic DNA scaffold MDC003775.57.

**Supplemental Figure S5.** Alignment of the MDP0000191488 (*MdGGP2*) consensus CDS from apple cultivars Braeburn (Br), Telamon (Tel), and Golden Delicious (GD).

**Supplemental Figure S6.** Alignment of the MDP0000191488 (*MdGGP3*) consensus CDS from apple cultivars Braeburn (Br), Telamon (Tel), and Golden Delicious (GD).

**Supplemental Figure S7.** Alignment of the MDP0000191488 (*MdGGP3*) consensus peptide sequences (translation products) from apple cultivars Braeburn (Br), Telamon (Tel), and Golden Delicious (GD).

**Supplemental Table S1.** Correlations between fruit antioxidant metabolite concentrations.

**Supplemental Table S2.** Correlations between fruit antioxidant metabolite concentrations over the three years of measurements.

**Supplemental Table S3.** Overview of population mean values and distribution for foliar AsA, DHA, totAsA, GSH, GSSG, and totGSH concentrations.

**Supplemental Table S4.** Correlations between GSH traits in foliar and in fruit flesh and skin tissues.

**Supplemental Table S5.** Correlations between fruit AsA concentrations and physiological traits.

**Supplemental Table S6.** QTL mapping for fruit antioxidant concentrations.

**Supplemental Table S7.** QTL mapping of fruit physiological traits.

**Supplemental Table S8.** Apple orthologs of candidate genes of AsA metabolism.

**Supplemental Table S9.** List of selected candidate gene orthologs of AsA metabolism mapped on the Telamon × Braeburn genetic linkage maps.

**Supplemental Table S10.** Activities of the clones of *Malus sylvestris* (GB CN915822) and apple (CN939721) with 100% identity to *MdGGP1* and *MdGGP2*, respectively, at the transcript level when expressed in *E. coli*.

**Supplemental Table S11.** Polymorphisms present in cv Braeburn and Telamon MDP0000172222 (*MdGGP1*) consensus CDS, and read coverage per allele.

**Supplemental Table S12.** Polymorphisms present in cv Braeburn and Telamon MDP0000288088 (*MdGGP2*) consensus CDS, and read coverage per allele.

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